```
function
ps_estimation(Btemp,Bdop,model,Nifgs,Narcs,Npsc_selections,std_param,sig2_est,br
eakpoint,breakpoint2,z)

% Function to estimate the ambiguities using LAMBDA and to
% calculate the residual phases
%
% Input:    - Btemp               temporal baselines [year]
%           - Bdop                Doppler baselines [Hz]
%           - model               vector with model indices to be
%                                 evaluated (see help model_definitions
%                                 for more information)
%           - Nifgs               number of interferograms
%           - Narcs               number of arcs
%           - Npsc_selections     number of psc selections
%           - std_param           standard deviations for pseudo-
%                                 observations
%           - sig2_est            estimated variance components
%           - breakpoint          breakpoint in case of double
%                                 linear model
%           - breakpoint2         second breakpoint, should be larger
%                                 than first breakpoint
%           - z                   current psc selection
%
% ------------------------------------------------------------------------
% File.............: ps_estimation.m
% Version & Date..: 1.7.2.16, 12-DEC-2009
% Author...........: Freek van Leijen
%                   Delft Institute of Earth Observation and Space Systems
%                   Delft University of Technology
% ------------------------------------------------------------------------
%
% This software is developed by Delft University of Technology and is
% intended for scientific use only. Applications for commercial use are
% prohibited.
%
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%
% Change log
%



% ------------------------------------------------------------------------
% Initialize
% ------------------------------------------------------------------------

global project_id ps_method

step1_orig = 1; % [m]
step2_orig = 0.0001; % [m], = 0.1 mm




% ------------------------------------------------------------------------
% Read data
% ------------------------------------------------------------------------

dpsc_fid = fopen([project_id '_dpsc_sel' num2str(z) '.raw'],'r');
dpsc_data = fread(dpsc_fid,[2*Nifgs+2 Narcs(z)],'double')';
fclose(dpsc_fid);

dpsc_phase = dpsc_data(:,3:Nifgs+2);
```

```
dpsc_h2ph = dpsc_data(:,Nifgs+3:2*Nifgs+2);
clear dpsc_data


% ----------------------------------------------------------------------
% Psc estimation
% ----------------------------------------------------------------------

switch ps_method
  case 'perio'

    [dpsc_param,dpsc_acheck,dpsc_model_info,dpsc_ens_coh] =
ps_periodogram_estimation(dpsc_h2ph,Btemp,dpsc_phase,model,std_param,step1_orig,
step2_orig);

    dpsc_varfac_best = repmat(NaN,Narcs(z),1);

  case {'ils','boot'}

    [dpsc_param,dpsc_acheck,dpsc_model_info,dpsc_varfac_best,dpsc_ens_coh] =
ps_ils_and_bootstrap_estimation(dpsc_h2ph,Btemp,Bdop,dpsc_phase,model,std_param,
sig2_est,breakpoint,breakpoint2);

    if isempty(find(~isnan(dpsc_model_info(:,1))));
      error('Something went wrong in the estimation of the network. Most likely
the stochastic model used is incorrect.');
    end

  otherwise

    error('Please select a valid ps method: ''period'', ''ils'' or ''boot''');

end

% ----------------------------------------------------------------------
% Write results to file
% ----------------------------------------------------------------------

dpsc_total = [dpsc_param dpsc_ens_coh dpsc_acheck dpsc_model_info
dpsc_varfac_best];
dpsc_fid = fopen([project_id '_dpsc_results_sel' num2str(z) '.raw'],'w');
fwrite(dpsc_fid,dpsc_total','double');
fclose(dpsc_fid);
```